

Govind Raj

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Professional Summary

Hands-on hardware and QA technician with 3+ years of experience in electronics lab environments, hardware diagnostics, and test automation. Skilled at supporting engineering teams through systematic troubleshooting, Python-based automation, and meticulous documentation. Strong foundation in embedded systems, PCBA-level testing, Linux/Windows environments, and cross-functional collaboration — with a track record of building tools that save measurable time and improve lab efficiency.

Core Competencies

- **Hardware & Lab Tools:** PCBA failure analysis, soldering/rework, oscilloscopes, multimeters, signal validation, schematic interpretation
- **Operating Systems:** Windows (advanced), Linux/Ubuntu (proficient), firmware/BIOS validation
- **Scripting & Automation:** Python, Selenium, Shell scripting — automated manual test workflows reducing cycle times by ~40%
- **Networking:** TCP/IP, DNS, DHCP — applied through embedded systems and firmware validation projects
- **Quality & Process:** ISO 9001, RCA, CAPA, NCR management, SOP documentation, 5S/Lean, Six Sigma Yellow Belt
- **Data & Reporting:** Excel dashboards, Power BI, defect tracking, FA documentation, calibration systems
- **AI-Assisted Dev Tools:** GitHub Copilot, Claude Code — code generation, refactoring, and debugging

Work Experience

Exact Technology Corporation — Quality Assurance Technician

Mar 2025 – Present

Toronto | Ontario | Canada

- Execute release and regression testing for engineering builds, validating hardware-software integration prior to production deployment.
- Perform root cause analysis and failure triage on complex system issues in close collaboration with firmware and BIOS engineering teams.
- Maintain test environments and equipment calibration schedules to ensure lab readiness for active engineering builds.
- Partner with cross-functional teams to drive issue resolution aligned with system performance and reliability targets.

Key Achievements:

- Automated manual regression testing workflows using Python and Selenium, cutting validation cycle time by ~40% across batches of 100+ hardware builds.
- Engineered a calibration lifecycle tracking system using Appsheet and Google Apps Script, integrating real-time email alerts and digital maintenance logs — eliminating manual audit prep and reducing compliance tracking time by an estimated 3–4 hours per audit cycle.
- Co-led the restart of the Health and Safety Committee; remains an active member driving ongoing compliance.

Crystal Fountains — Quality Assurance Technician

Nov 2021 – Mar 2025

Vaughan | Ontario | Canada

- Conducted end-to-end validation and debug of PCBA assemblies and control systems for embedded electronics products.
- Managed non-conformance reports through detailed failure analysis and defect categorization to isolate firmware and hardware root causes.
- Maintained comprehensive quality records, FA reports, and audit documentation ensuring full ISO 9001 traceability.

- Operated lab instruments including oscilloscopes, thermal analyzers, and power supplies for signal validation and component-level debug.
- Coordinated lab reorganization, equipment moves, and workspace setup as part of ongoing lean optimization initiatives.

Key Achievements:

- Led 5S implementation across 3 QA lab testing zones, reducing equipment retrieval time by ~30% and enabling faster test setup — increasing daily testing throughput measurably.
- Redesigned the RMA intake and testing workflow, standardizing defect categorization and cutting average processing time by ~25% while improving first-pass resolution accuracy.
- Partnered with R&D teams to develop updated testing methods aligned with evolving product requirements and organizational best practices.

Advanced Skytech Canada—Hardware Tester

Mar 2021–Nov 2021

Markham | Ontario | Canada

- Performed hardware diagnostics, soldering/rework, and visual inspection of PCBAs and electronic sub-assemblies.
- Interpreted complex schematics and datasheets to identify faulty components and verify circuit compliance.
- Operated test instruments — oscilloscopes, thermal analyzers, power supplies — for signal validation and component-level debug.
- Documented failure patterns and rework procedures to support continuous process improvement across production lines.

Certifications & Licenses

- **Six Sigma Yellow Belt Certification**—Council of Six Sigma Certification
- **Working at Heights** – LIFT Training
- **Quality Management Foundations** – LinkedIn Learning
- **Excel Essential Trainings (Microsoft 365)** – LinkedIn Learning
- **IPC A 610 Operators Training Program**

Education

Post-Graduate Diploma, Embedded Systems Engineering Design —Lambton College, Toronto, Canada
Bachelor of Engineering, Electronics & Communication Engineering —Mahatma Gandhi University, India.